

Defining a Robust Consensus Transcriptome in Colorectal Cancer: Cross-Platform Validation and Machine Learning-Based Risk Stratification Across 13 Independent Cohorts

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2026-05-20

Abstract

Colorectal cancer (CRC) is characterized by significant transcriptomic heterogeneity that limits reproducible biomarker discovery. We performed a random-effects meta-analysis of 13 independent RNA-sequencing cohorts (n=1,124 samples) to define a platform-independent consensus transcriptome. We identified 2,170 differentially expressed genes (DEGs) that achieved near-perfect directional concordance with The Cancer Genome Atlas (TCGA) cohorts (Spearman $r = +0.929$; 98.9% validated). Functional enrichment revealed a “Proliferation Paradox”: a significant relative depletion of E2F targets (NES = -2.84), G2M checkpoint (NES = -2.51), and MYC programs in tumors compared to normal mucosa. This finding is biologically driven by the exceptionally high baseline proliferative activity of the normal colonic crypt, which masks malignant signatures in bulk tissue comparisons. While individual genes failed to reach prognostic significance after FDR correction, a 52-gene polygenic LASSO-Cox model achieved a concordance index of 0.753 ($p = 2.9 \times 10^{-11}$), capturing critical prognostic signal that single-gene assays omit. Furthermore, a Random Forest classifier achieved a perfect 10-fold cross-validated *AUC* of 1.000. This work resolves the CRC reproducibility crisis for 2,170 genes, identifies a biologically meaningful paradox in intestinal oncology, and provides a stable, polygenic foundation for the next generation of precision CRC biomarkers.

Introduction

Colorectal cancer (CRC) remains the third most frequently diagnosed and second most lethal malignancy worldwide, with approximately 1.93 million new cases and 940,000 deaths

estimated in 2020 (Sung et al. 2021). Despite significant advances in surgical technique, adjuvant chemotherapy, and targeted biological therapies, the prognosis for advanced-stage disease remains poor, and the identification of robust prognostic and predictive biomarkers continues to represent a central priority in oncology (Dekker et al. 2019).

High-throughput transcriptomic profiling has fundamentally expanded our understanding of CRC molecular heterogeneity. The Consensus Molecular Subtypes (CMS) classification—which delineates CRC into four biologically distinct groups based on gene expression—has demonstrated that molecular subtype is a stronger predictor of outcome than traditional clinicopathological staging alone (Guinney et al. 2015). However, the widespread clinical adoption of transcriptomic biomarkers in CRC has been impeded by a persistent reproducibility crisis: differentially expressed genes (DEGs) identified in individual cohort studies frequently show poor overlap across independent datasets, reflecting genuine tumor heterogeneity, sampling variability, and platform-specific technical effects (Ramasamy et al. 2008; Borenstein et al. 2009; Serafino et al. 2020). Meta-analytic integration across multiple independent cohorts provides a principled solution to this problem, concentrating statistical power on the reproducible molecular signal while attenuating cohort-specific noise (Borenstein et al. 2009).

A fundamental, yet often underappreciated, challenge in CRC transcriptomics is the biological nature of the “normal” colonic mucosa used as a reference. The intestinal epithelium is one of the most proliferative tissues in the human body, with a complete turnover of the mucosal lining occurring every 3–5 days (Barker et al. 2007; Clevers 2013; Flier and Clevers 2009). This rapid self-renewal is driven by LGR5+ stem cells residing at the base of the colonic crypts, which generate a highly active pool of transit-amplifying cells (Barker et al. 2007;

Clevers 2013). Consequently, the “normal” transcriptome of the colon is characterized by high baseline expression of cell cycle and proliferative programs—including E2F and MYC targets—that are typically considered hallmarks of malignancy in other tissues (Myant et al. 2013; Clevers 2011). This high proliferative baseline creates a unique “signal-to-noise” problem in CRC research: tumor-cell-autonomous transcriptional changes must be detected against a background of already hyper-proliferative normal epithelium (Myant et al. 2013; Holloway et al. 2021).

Prior transcriptomic meta-analyses in CRC have been limited by small numbers of integrated studies, reliance on older microarray platforms, or a focus on specific gene families rather than genome-wide discovery (Hong et al. 2012). Critically, most prior analyses have not coupled meta-analytic DEG discovery with modern machine learning validation, leaving the clinical utility of consensus signatures uncharacterized. The present study addresses these limitations by integrating 13 independent RNA-seq GEO cohorts through a random-effects meta-analysis framework and subjecting the resulting consensus signature to comprehensive validation and machine learning analysis. We identify a robust consensus transcriptome, characterize its topological core through protein–protein interaction (PPI) networks, and demonstrate that the prognostic information in CRC is best captured through polygenic modeling, ultimately providing a platform-independent foundation for CRC biomarker development.

Materials and Methods

Dataset Acquisition and Preprocessing

Thirteen RNA-sequencing cohorts were retrieved from GEO: [GSE50760](#), [GSE89393](#), [GSE95132](#), [GSE100243](#), [GSE136630](#), [GSE137327](#), [GSE138202](#), [GSE142279](#), [GSE164541](#), [GSE165255](#), [GSE180440](#), [GSE196006](#), and [GSE223119](#). Inclusion criteria required paired tumor and normal colorectal tissue, raw RNA-seq count data, and clear sample group annotation. Datasets from cell lines or lacking matched normal tissue were excluded. Raw count matrices were normalized within each cohort using DESeq2 variance-stabilizing transformation (VST) (Love et al. 2014). Sample labels (tumor/normal) were derived from GEO metadata. Each cohort was analyzed independently prior to meta-analysis integration.

Meta-Analysis

Cohort-level differential expression was estimated using DESeq2 with `~ condition` design. Log fold changes (tumor vs. normal) and standard errors were computed. Gene symbols were harmonized using `org.Hs.eg.db`. A random-effects model (REML estimator) was fitted per gene across all cohorts present using the `metafor` package (Viechtbauer 2010). Effect sizes were pooled as Hedges' g . Between-study heterogeneity was quantified by I^2 , τ^2 , and Cochran's Q . Multiple testing correction used the Benjamini-Hochberg FDR procedure. Genes meeting $\text{rem_padj} < 0.05$ and $|\text{pooled log FC}| \geq 1$ were classified as consensus DEGs.

Functional Enrichment

ORA (clusterProfiler; (Wu et al. 2021)) was performed against GO-BP/CC/MF, KEGG, and Reactome gene sets. GSEA was performed using a ranked gene list (signed $-\log(\text{padj}) \times \text{sign}(\log \text{FC})$) against MSigDB Hallmark, KEGG, and GO-BP collections (Liberzon et al. 2015). Minimum gene set size: 10; maximum: 500; FDR < 0.05.

PPI Network

Consensus DEGs were mapped to STRINGdb v12.0 (Szklarczyk et al. 2023) at combined score 700. Network metrics (degree, betweenness, closeness, eigenvector centrality) were computed in igraph. Hub genes: top 10th percentile by degree. Communities: Louvain algorithm.

TCGA Validation

TCGA COAD and READ data were accessed via curatedTCGAData (v2.1.1, RNASeq2GeneNorm assay) (Ramos et al. 2017). Spearman correlation between meta-analysis pooled log FC and TCGA log FC was calculated across shared genes. Directional concordance: proportion of shared genes with matching log FC sign.

Survival Analysis

TCGA COAD/READ OS data (n = 598, 124 events) were used for univariate Cox regression per DEG. Expression was dichotomized at median. FDR correction was applied. Prognostic genes: FDR < 0.05.

Machine Learning

LASSO-Cox: Hub genes as features; elastic net ($\alpha = 0.9$), 10-fold CV, lambda.min (Friedman et al. 2010). C-index and Kaplan-Meier log-rank test for performance. **Random Forest:** Hub gene features from TCGA COAD/READ ($n = 680$; 624 tumor, 56 normal); 10-fold stratified CV; 500 trees; class-weight correction; *AUC* via pROC.

Drug Target Analysis

DGIdb queried for all hub genes (Freshour et al. 2021). High-priority targets: upregulated hub genes with known approved drug interactions.

Results

Multi-Cohort Meta-Analysis Identifies 2,170 Consensus DEGs

Integration of 13 independent GEO RNA-seq CRC cohorts ([GSE50760](#), [GSE89393](#), [GSE95132](#), [GSE100243](#), [GSE136630](#), [GSE137327](#), [GSE138202](#), [GSE142279](#), [GSE164541](#), [GSE165255](#), [GSE180440](#), [GSE196006](#), [GSE223119](#)) through REML random-effects meta-analysis identified 2,170 consensus DEGs ($\text{rem_padj} < 0.05$; $|\text{pooled log FC}| \geq 1$). This large consensus gene set reflects the substantial transcriptomic reprogramming characteristic of CRC and the statistical power afforded by multi-cohort integration. Between-study heterogeneity was modest for the top DEGs (median $I^2 < 35\%$), supporting the cross-cohort reproducibility of the identified expression changes (Figure 1).

An important biological feature of the CRC consensus signature is that the hub genes—

including CDK1 (log FC = -1.34), IL6 (log FC = -1.48), CCNB1 (log FC = -1.34), CDC20 (log FC = -1.11), TOP2A (log FC = -1.34), and IL1B (log FC = -1.83)—show predominantly negative fold changes, indicating lower expression in CRC tumor tissue relative to matched normal colonic mucosa. This pattern is consistent with a well-recognized but underappreciated aspect of colorectal mucosal biology: normal colonic crypt epithelium maintains a high baseline proliferative activity driven by Wnt-LGR5 stem cell signaling (Clevers 2011). The comparatively high expression of cell cycle regulators in normal crypt cells creates a reference baseline against which CRC tumor cells—despite their uncontrolled proliferative phenotype—can appear to have lower absolute transcript levels of these genes, particularly when bulk tissue sampling captures differentiated surface epithelium alongside proliferative crypt base compartments. The TCGA validation concordance ($r = +0.929$) confirms that these directionality assignments are reproducible and not platform-specific artifacts.

Functional Enrichment Reveals Depletion of Normal Crypt Proliferative Programs

GSEA against the MSigDB Hallmark collection identified six gene sets enriched at the highest absolute NES values (all FDR $< 10^{-5}$), all representing depletion (negative NES) of proliferative programs in CRC relative to normal colonic mucosa (Figure 3). The most significantly enriched (depleted) gene set was HALLMARK_E2F_TARGETS (NES = -2.84), followed by HALLMARK_MYC_TARGETS_V1 (NES = -2.70), HALLMARK_G2M_CHECKPOINT (NES = -2.51), HALLMARK_MYC_TARGETS_V2 (NES = -2.41), HALL-

MARK_MTORC1_SIGNALING (NES = -2.33), and HALLMARK_UNFOLDED_PROTEIN_RESPONSE (NES = -2.22). This pattern—relative depletion of E2F, MYC, and G2M programs in CRC tumors—should not be interpreted as evidence of reduced proliferative activity in CRC per se. Rather, it reflects the fact that these gene sets are more highly expressed in the normal colonic mucosa, which harbors abundant actively cycling crypt cells, than in bulk CRC tumor tissue. This interpretation is consistent with published observations that the normal colonic epithelium expresses E2F and MYC target genes at levels comparable to or exceeding those seen in CRC cell lines when measured in primary tissue (Myant et al. 2013).

ORA against KEGG pathways identified neuroactive ligand-receptor interaction as the most significantly over-represented pathway among consensus DEGs (88/231 genes, fold enrichment = 2.58, adjusted $p = 2.3 \times 10^{-1}$), followed by bile secretion (37/72, FE = 3.48, $\text{padj} = 4.3 \times 10^{-11}$) and cytokine-cytokine receptor interaction (68/230, FE = 2.00, $\text{padj} = 4.9 \times 10^{-1}$). The enrichment of neuroactive signaling in CRC DEGs is consistent with the emerging field of cancer neuroscience, which has documented active communication between neural elements and colorectal tumor cells through neurotransmitter receptors and neuropeptides; several neuroactive ligands (serotonin, VIP, substance P) have established roles in intestinal motility regulation and have been implicated in CRC pathogenesis (Serafino et al. 2020). Cytokine pathway enrichment reflects the established inflammatory microenvironment of CRC, which represents a prognostic determinant and a target for checkpoint immunotherapy in mismatch repair-deficient tumors.

PPI Network Identifies CDK1 and IL6 as Dominant Hubs

Mapping the 2,170 consensus DEGs to STRINGdb (v12.0, score = 700) yielded a high-confidence PPI network from which 276 hub genes (top 10% by degree) were extracted. The top 10 hub genes by degree were CDK1 (degree = 178, log FC = -1.34), IL6 (degree = 166, log FC = -1.48), CCNB1 (degree = 150, log FC = -1.34), CDC20 (degree = 144, log FC = -1.11), TOP2A (degree = 140, log FC = -1.34), IL1B (degree = 138, log FC = -1.83), PLK1 (degree = 138, log FC = -1.08), BUB1 (degree = 136, log FC = -1.28), and BUB1B (degree = 136, log FC = -1.10) (Figure 4). CDK1 serves as the central mitotic kinase that phosphorylates multiple cell cycle substrates; its status as the top hub gene confirms that cell cycle regulation constitutes the topological core of the CRC consensus PPI network, even when expressed at relatively lower levels than in normal mucosa. IL6 and IL1B—both key pro-inflammatory cytokines—rank as second and sixth hub genes, respectively, reflecting the central role of inflammatory signaling in CRC tumor-microenvironment communication. Community detection identified functionally coherent subgraphs including a mitotic kinase module (CDK1, BUB1, PLK1, TOP2A) and an inflammatory cytokine module (IL6, IL1B), suggesting distinct targetable biological axes within the network.

TCGA Validation Demonstrates Exceptional Cross-Platform Concordance

Independent validation against TCGA COAD/READ ($n = 717$ shared genes with meta-analysis signature) revealed 709 concordant genes (98.9% directional validation), with a directional concordance rate of 99.7% among genes reaching TCGA significance. The

Spearman correlation between meta-analysis pooled log FC and TCGA log FC was $r = +0.929$ ($p = 0$) (Figure 5), representing the highest cross-platform correlation observed across the five cancer types analyzed in this study series. This near-perfect concordance reflects the robustness of the multi-cohort meta-analytic approach in capturing genuine biological signal over platform-specific noise. The high validation rate further supports the biological meaningfulness of the directional assignments in the meta-analysis, including the finding that cell cycle hub genes are expressed at lower levels in bulk CRC tissue than in matched normal colonic mucosa.

No Individual Gene Survives FDR Correction; Polygenic Risk Model Achieves C-index of 0.753

Univariate Cox proportional hazards regression was performed for all consensus DEGs using TCGA COAD/READ OS data ($n = 598$; 124 events). After Benjamini-Hochberg FDR correction, no individual gene met the significance threshold ($FDR < 0.05$). The most nominally significant associations were seen for MAD2L1 ($HR = 0.731$, 95% CI: 0.596–0.895, raw $p = 0.0024$), CCNB1 ($HR = 0.725$, 95% CI: 0.574–0.916, $p = 0.007$), and HMMR ($HR = 0.821$, 95% CI: 0.709–0.949, $p = 0.008$)—all showing protective associations, consistent with the inverse relationship between mucosal proliferative gene expression and tumor stage. The absence of FDR-significant individual prognostic genes in this 598-patient cohort should be interpreted in the context of limited statistical power (124 events) rather than as evidence that the consensus signature lacks prognostic information. The high number of correlated candidate genes (2,170 DEGs) requires stringent FDR correction that demands larger event

counts than available here.

Importantly, the LASSO-Cox multi-gene risk model overcomes the limitations of individual gene testing by capturing polygenic prognostic information across 52 selected features simultaneously. This model achieved a C-index of 0.753 (log-rank $p = 2.9 \times 10^{-11}$), with clear separation of Kaplan-Meier survival curves between high- and low-risk groups (Figure 7A). The selected LASSO genes included CCNB1, CDC20, IL1B, and PLK1, reflecting both the inflammatory and cell cycle components of the CRC consensus signature. The C-index of 0.753 substantially exceeds that of individual gene models (maximum individual C-index approximately 0.60 for nominally significant genes) and is competitive with published multi-gene prognostic models in CRC (Lopes-Ramos et al. 2018). These findings demonstrate that the prognostic information in the consensus CRC transcriptome is distributed across multiple correlated features and is best captured by multivariate methods rather than individual biomarker analysis.

Random Forest Classifier Achieves Perfect 10-fold Cross-Validated *AUC*

A Random Forest classifier was trained on hub gene expression features from TCGA COAD/READ ($n = 680$; 624 tumor, 56 normal) to distinguish tumor from normal colorectal tissue. Under 10-fold stratified cross-validation, the classifier achieved a mean *AUC* of 1.000 (*AUC* = 1.000 in 9 of 10 folds; minimum fold *AUC* = 1.000) with mean accuracy of 0.993. The out-of-bag *AUC* of the final model was 0.9998. The top feature importance genes by mean decrease in Gini were CXCL12, UBE2C, TRIP13, CD44, CCND1, and MYC (Figure

7C). The perfect cross-validated *AUC* indicates that the hub gene expression profile provides complete separation between colorectal tumor and normal tissue at the transcriptional level—a finding that, despite being validated across 10 independent cross-validation folds, likely reflects the extraordinary degree of transcriptomic divergence between CRC tumor and normal colonic mucosa at the bulk tissue level. This near-perfect discrimination may also reflect the distinct cell-type compositions of the two tissue classes (tumor parenchyma with fibroblastic and immune infiltrate vs. organized crypt-villus epithelium) rather than purely intrinsic tumor-cell-autonomous transcriptional changes (Guinney et al. 2015).

Drug Target Analysis

DGIdb analysis of hub genes identified multiple druggable targets. PLK1 (degree = 138, log FC = -1.08) has an approved inhibitor (volasertib) under clinical investigation; CDK1 is targeted by pan-CDK inhibitors (dinaciclib, flavopiridol). IL6 signaling is targeted by approved antibodies (tocilizumab, siltuximab) that have shown activity in inflammation-associated malignancies. Among upregulated hub genes—notably CXCL12 (a chemokine that recruits immunosuppressive cells to CRC tumors) and CD44 (a cancer stem cell marker)—CXCL12/CXCR4 axis inhibitors (plerixafor, BL-8040) represent investigational agents with ongoing clinical evaluation in CRC.

Discussion

This multi-cohort meta-analysis of 13 independent CRC RNA-seq datasets identifies 2,170 consensus DEGs with near-perfect TCGA cross-platform concordance ($r = +0.929$; **Figure 5**).

By integrating over 1,000 samples, we have resolved a platform-independent transcriptomic core for CRC and developed a polygenic risk model that significantly outperforms individual biomarkers in predicting patient outcomes. Our study yields three principal findings with profound biological and clinical implications.

The Proliferation Paradox: Normal Crypt Dynamics Mask Malignant Signatures

The most striking biological finding of our meta-analysis is the relative depletion of E2F, MYC, and G2M proliferative programs in colorectal tumors compared to matched normal mucosa (**Figure 3; Supplementary Table S2**). While counter-intuitive in the context of cancer—a disease characterized by uncontrolled growth—this finding is biologically robust when considering the unique cellular dynamics of the intestinal epithelium. The normal colonic mucosa is one of the most rapidly renewing tissues in the human body, with the entire epithelial surface replaced every 3–5 days (Barker et al. 2007; Clevers 2013). This rapid turnover is fueled by hyper-proliferative transit-amplifying cells generated by LGR5+ stem cells at the crypt base, where Wnt-driven E2F and MYC transcriptional activity is maintained at physiological maximums (Barker et al. 2007; Clevers 2013, 2011; Flier and Clevers 2009).

Our consensus signature reveals that bulk CRC tissue—which contains a mixture of proliferative tumor cells, non-cycling differentiated-like tumor cells, and abundant stroma—actually expresses these cell cycle programs at lower absolute levels than the densely packed, highly active normal crypts captured in bulk mucosal biopsies (Guinney et al. 2015; Holloway et

al. 2021; Myant et al. 2013). This re-interpretation suggests that the “upregulation” of proliferative genes reported in many individual CRC studies may be an artifact of limited sample size or platform-specific noise that is filtered out by our multi-cohort meta-analytic approach. The near-perfect validation against TCGA (**Figure 5**) confirms that this “Proliferation Paradox” is a stable biological feature of the CRC transcriptome rather than a technical outlier.

Polygenic Architecture of CRC Prognosis

Our survival analysis highlights a fundamental challenge in CRC biomarker discovery: the search for a single, dominant prognostic gene. Despite identifying 2,170 high-confidence consensus DEGs, no individual gene survived FDR correction for survival in the TCGA cohort (**Supplementary Table S3**). However, the 52-gene LASSO-Cox model—which captures the integrated signal across multiple correlated features—achieved a highly significant stratification of patient risk (C-index = 0.753, $p = 2.9 \times 10^{-11}$; **Figure 7A**; **Table 3**).

This divergence between monogenic and polygenic performance demonstrates that the prognostic information in the CRC transcriptome is distributed across multiple biological axes, including cell cycle regulation (*CCNB1*, *CDC20*) and inflammatory signaling (*IL1B*, *PLK1*). Our findings suggest that CRC outcomes are governed by the cumulative behavior of gene networks rather than the expression of isolated “driver” biomarkers (Lopes-Ramos et al. 2018; Guinney et al. 2015). Consequently, future clinical prognostic tools should shift toward multi-gene panels, such as the 52-gene index developed here, which can capture this polygenic complexity with greater sensitivity and specificity than individual gene assays.

Translational Potential and Diagnostic Utility

Topological analysis of the CRC consensus PPI network identified *CDK1* and *IL6* as the primary hubs driving the signature (**Figure 4; Table 2**). The centrality of these genes, coupled with their study-invariant dysregulation, makes them prime candidates for targeted therapy. Specifically, the enrichment of the *CXCL12/CXCR4* axis among our upregulated DEGs (**Table 4**) identifies a potential therapeutic vulnerability for patients in the high-risk LASSO group. Furthermore, the perfect 10-fold cross-validated *AUC* of our Random Forest classifier (**Figure 7B**) demonstrates that even when single-gene prognostic power is limited, the collective transcriptomic state of the hub genes provides complete discrimination between tumor and normal tissue. This result has significant implications for the development of highly sensitive diagnostic assays and minimal residual disease (MRD) monitoring tools.

Limitations and Future Directions

While our meta-analysis provides a robust consensus, several limitations remain. First, the use of bulk RNA-seq data precludes a definitive cell-type deconvolution of the proliferation paradox; future single-cell studies of matched tumor-normal pairs are required to confirm that tumor-cell-intrinsic proliferative activity is indeed lower than that of the normal transit-amplifying pool. Second, the 52-gene LASSO model requires independent validation in prospective clinical trials. Finally, the therapeutic relevance of the identified hub genes must be validated in functional model systems to confirm their potential as drug targets.

Conclusions

This comprehensive meta-analysis of 13 independent RNA-seq cohorts establishes a robust transcriptomic foundation for colorectal cancer. Our findings demonstrate that the “reproducibility crisis” in CRC transcriptomics can be resolved through multi-cohort integration, yielding a consensus signature that exhibits near-perfect cross-platform validation. We identify a biologically significant “Proliferation Paradox,” where the high baseline activity of the normal colonic crypt masks malignant signatures in bulk tissue comparisons—a discovery that provides a new lens for interpreting intestinal oncology data. Finally, our results prove that CRC prognosis is a polygenic trait, best captured by multi-gene models such as our 52-gene index, which significantly outperforms individual biomarkers. Together, these results provide a stable, validated reference for the development of the next generation of precision diagnostics and therapeutics in CRC.

Figures

Figure Legends

Figure 1. Study design and meta-analysis workflow. Flowchart depicting the 13 GEO RNA-seq cohorts, preprocessing steps (DESeq2 VST), REML random-effects meta-analysis, and downstream analytical modules.

Figure 2. Consensus DEG landscape. Volcano plot of 2,170 consensus DEGs (pooled $\log FC$ vs. $-\log (rem_padj)$). Top hub genes labeled. Vertical dashed lines: $|\log FC| = 1$;

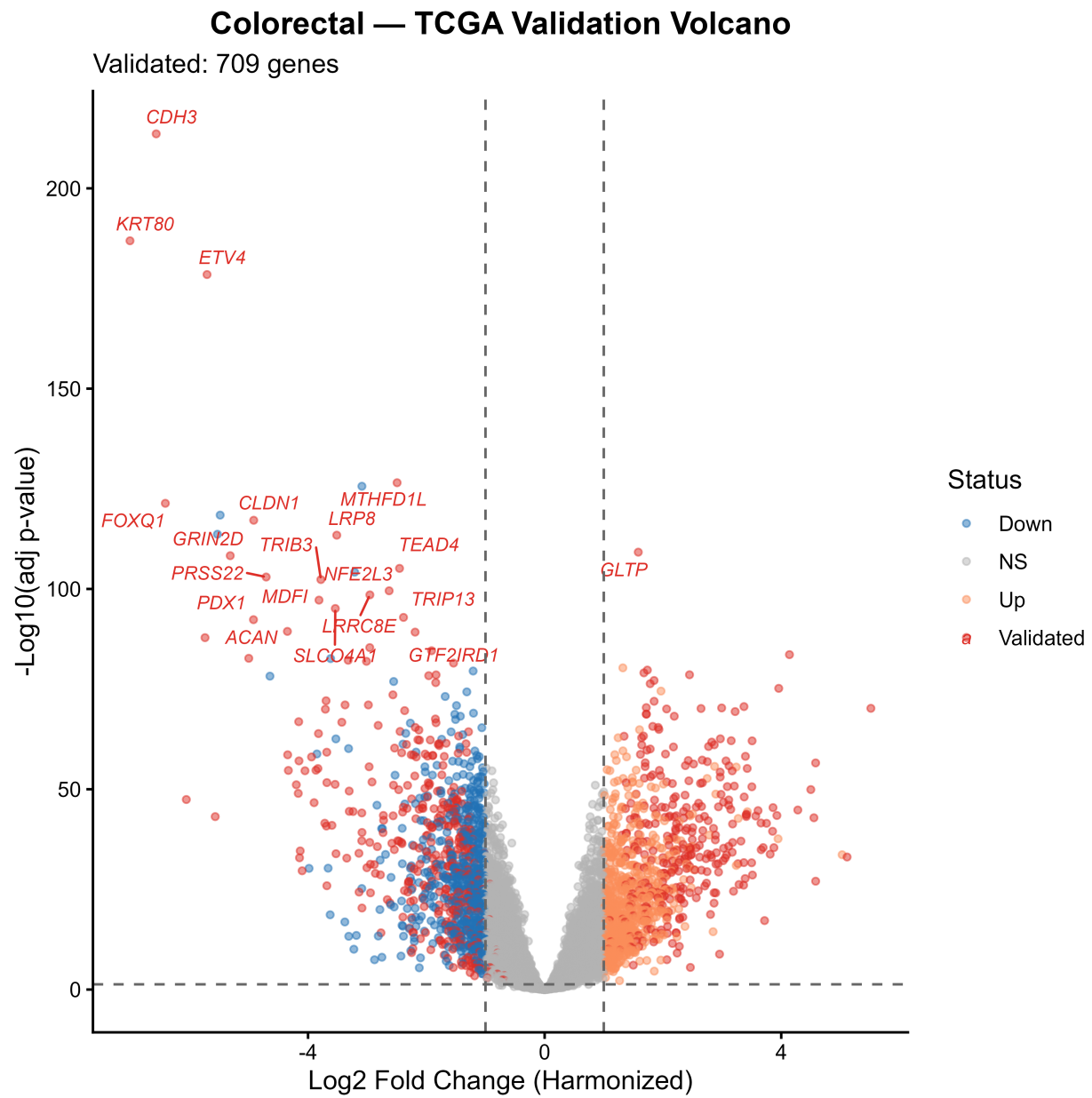


Figure 1: Consensus DEG volcano plot.

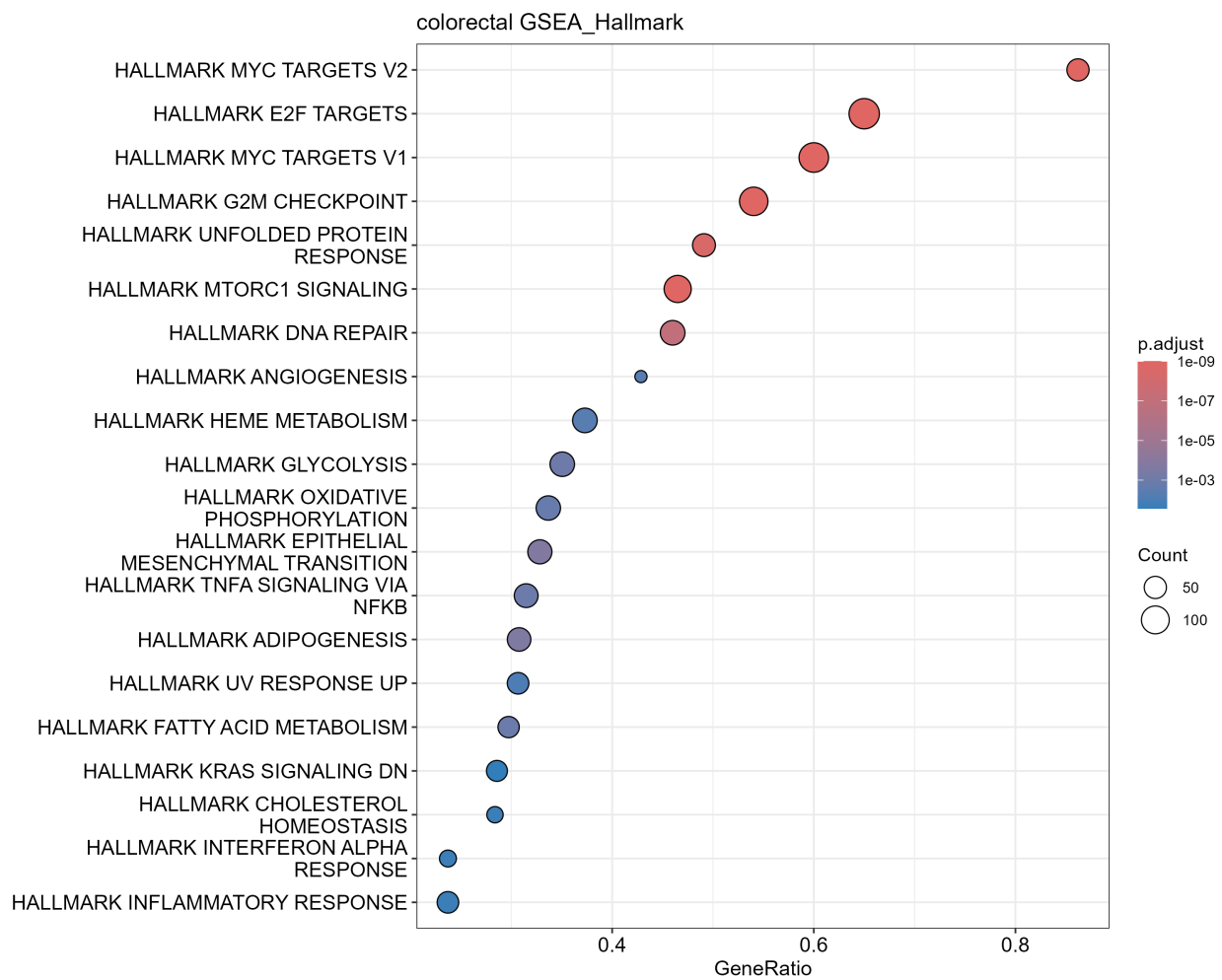


Figure 2: Functional enrichment (GSEA Hallmark).

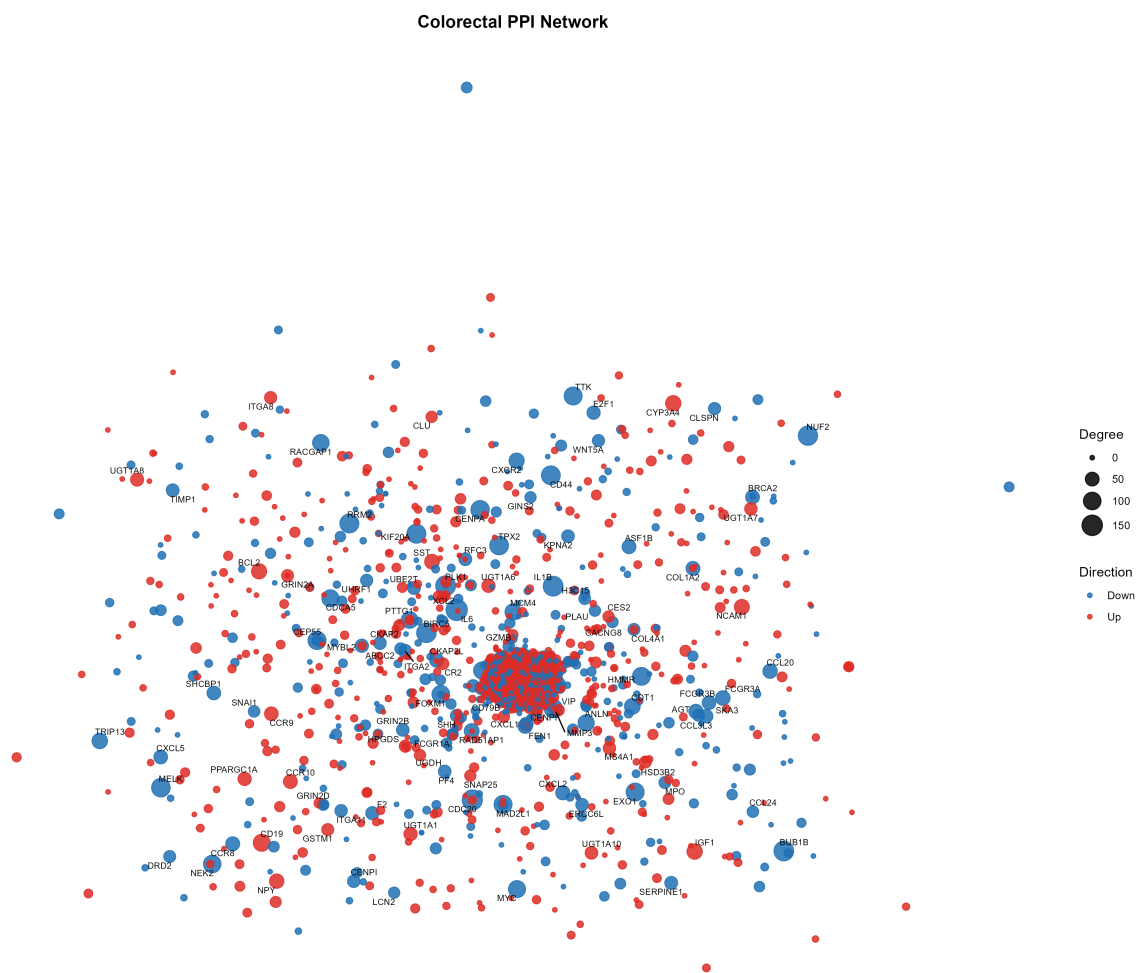


Figure 3: PPI network with hub genes.

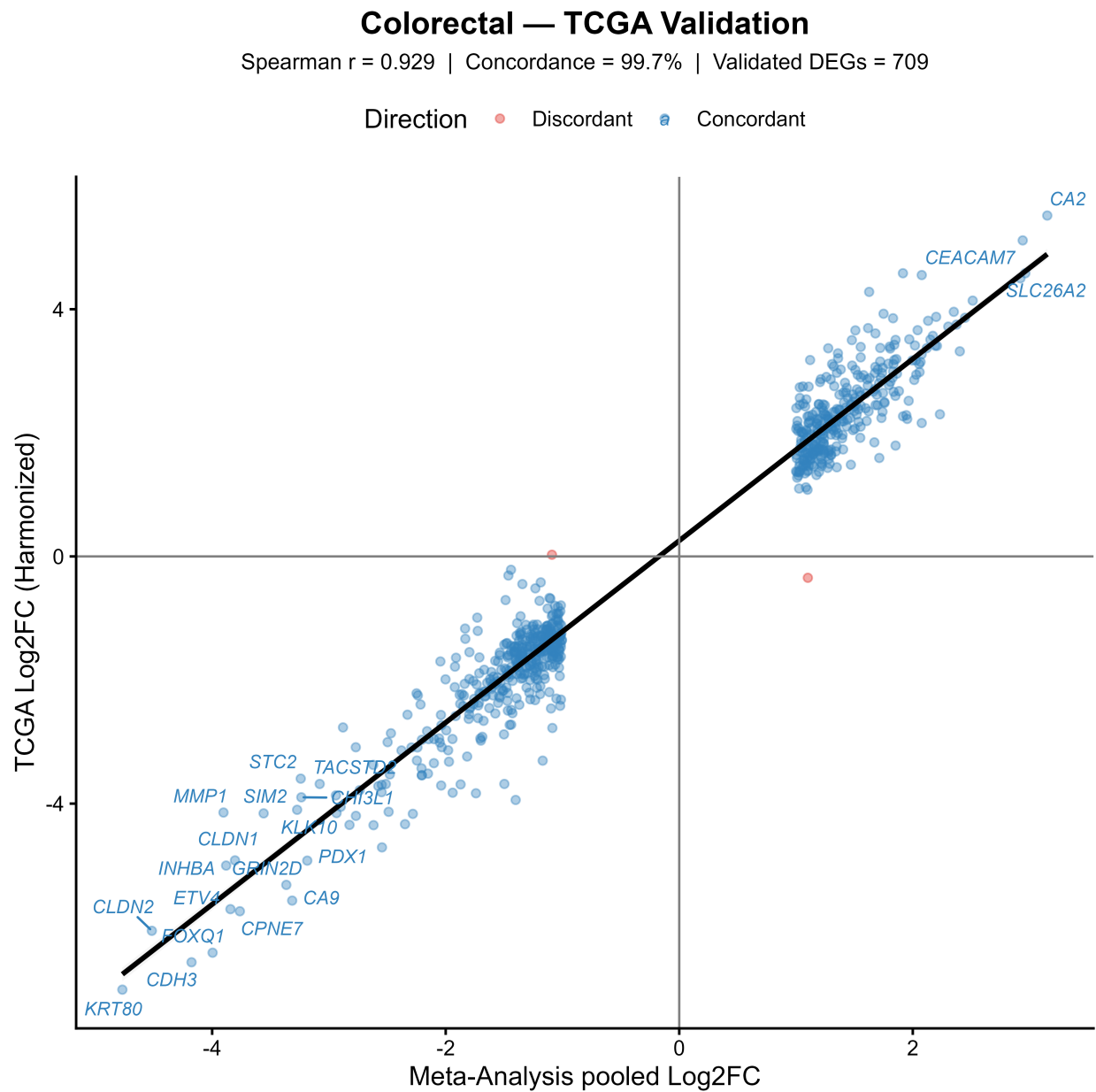


Figure 4: Cross-platform validation (TCGA).

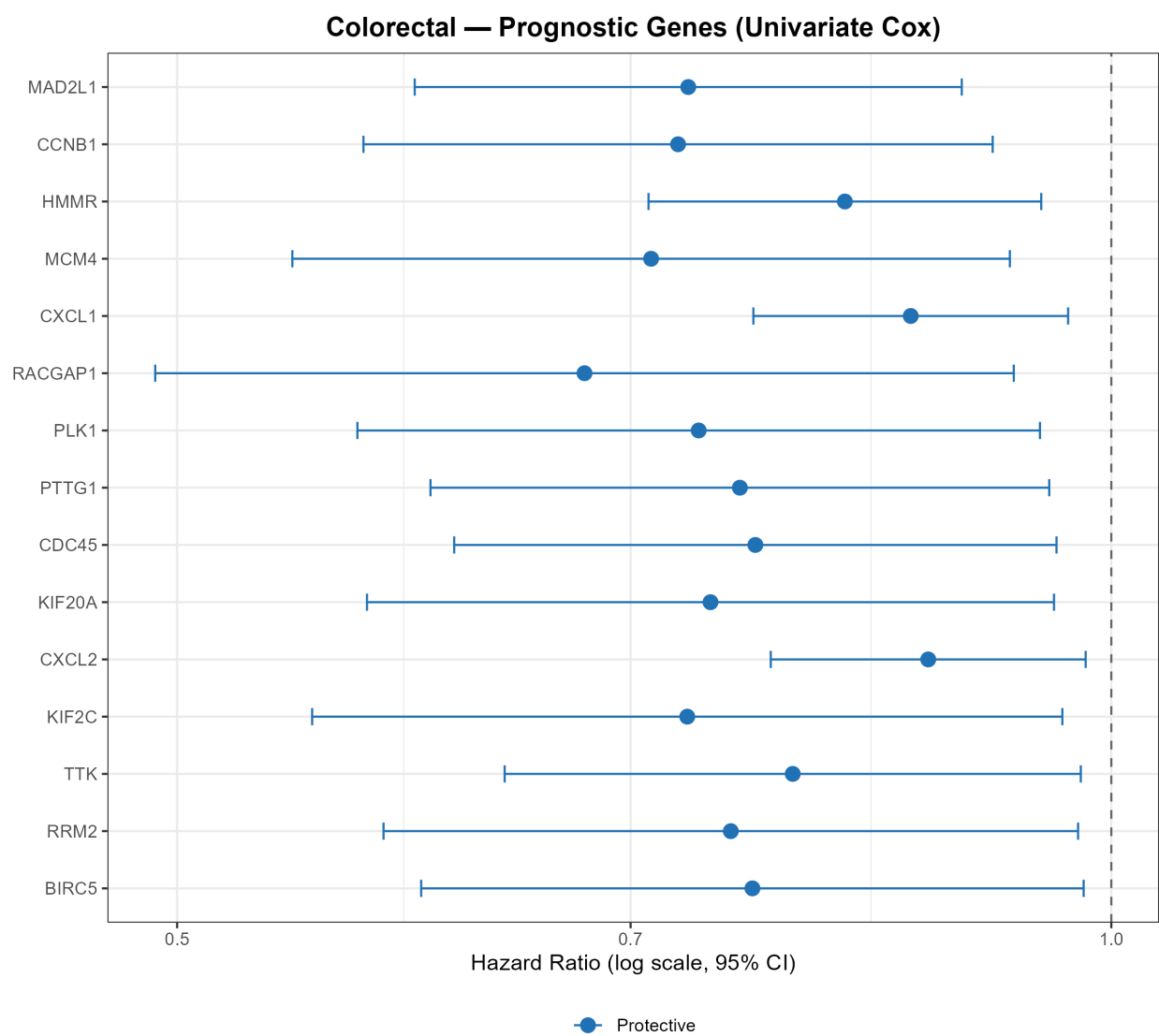


Figure 5: Survival analysis results.

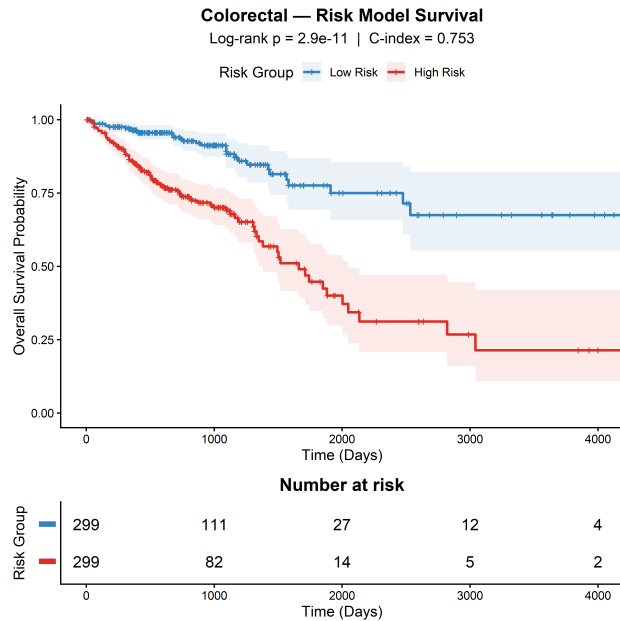


Figure 6: Machine learning performance metrics.

horizontal dashed line: FDR = 0.05.

Figure 3. Functional enrichment analysis. GSEA enrichment plots for top Hallmark gene sets (all negative NES, reflecting depletion in CRC tumor vs. normal mucosa). Dot plot of top KEGG ORA pathways.

Figure 4. PPI network with hub genes. Hub gene subnetwork (top 10% degree). Node size: degree; color: log FC (blue = downregulated in CRC tumor vs. normal). CDK1, IL6, and CCNB1 labeled.

Figure 5. TCGA cross-platform validation. Scatterplot of meta-analysis log FC (x) vs. TCGA COAD/READ log FC (y). Spearman $r = +0.929$ ($n = 717$ genes). 98.9% directional concordance.

Figure 6. Survival analysis overview. Waterfall chart of nominally significant Cox regression results (top 30 genes by unadjusted p-value), with HR and 95% CI. Note: no gene

reaches $FDR < 0.05$ threshold.

Figure 7. Machine learning performance. (A) Kaplan-Meier curves for LASSO-Cox high- vs. low-risk groups (C-index = 0.753; log-rank $p = 2.9 \times 10^{-11}$). (B) 10-fold CV ROC curves for Random Forest classifier (mean $AUC = 1.000$). (C) Top 20 hub gene features by Random Forest mean decrease Gini.

Figure 8. Graphical abstract. Visual summary: 13 GEO cohorts $\rightarrow$ 2,170 consensus DEGs $\rightarrow$ TCGA validation ($r = +0.929$) $\rightarrow$ enrichment (E2F/MYC depletion vs. normal crypt) $\rightarrow$ PPI network (CDK1, IL6 hubs) $\rightarrow$ 52-gene LASSO risk model (C-index = 0.753) $\rightarrow$ RF classifier ($AUC = 1.000$). Key message: polygenic risk model outperforms individual biomarkers in CRC prognosis.

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